An Exact Decomposition of the Divisor Summatory Function and Fixed-j Asymptotics

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Abstract. In this paper, we study a decomposition of the divisor summatory function $D(x)$. We establish the exact identity

$$D(x) = 1 + \sum_{i \geq 1} (i+1)\pi(x^{1/i}) + \sum_{j \geq 2} (-1)^j (j-1) T_j(x),$$

which decomposes $D(x)$ via inclusion–exclusion according to the number of distinct prime factors of n. The three terms correspond respectively to $n = 1$, prime powers, and integers with $\omega(n) \geq 2$.

We then establish an asymptotic formula with a stated error term for $T_j(x)$ with fixed $j \geq 2$. We consider the two-variable Dirichlet series $F(s, y)$ and obtain the factorization $F(s, y) = \zeta(s)^{2+2y} H(s, y)$, where H is holomorphic near $s = 1$. Applying the Selberg–Delange method—a truncated Perron formula, contour deformation, and local Hankel analysis—we prove that

$$T_j(x) \sim \frac{2^j}{j!}\, x \log x \,(\log\log x)^j \quad \text{as } x \to \infty.$$

These estimates describe the individual fixed-j components of the decomposition, rather than the full alternating sum.

1 Introduction

1.1 The Dirichlet divisor problem

Let $d(n)$ denote the number of divisors of a positive integer n, and let

$$D(x) := \sum_{n \leq x} d(n) \tag{1.1}$$

be its summatory function. Dirichlet [2] proved in 1849 that

$$D(x) = x \log x + (2\gamma - 1)x + \Delta(x), \tag{1.2}$$

where γ is the Euler–Mascheroni constant and $\Delta(x) = O(x^{1/2})$. The Dirichlet divisor problem asks for the infimum θ of all real numbers such that $\Delta(x) = O(x^{\theta+\varepsilon})$ for every $\varepsilon > 0$.

Hardy [3] proved that $\theta \geq 1/4$, and it is conjectured that $\theta = 1/4$. In the other direction, Voronoï [6] proved the classical estimate

$$\Delta(x) = O(x^{1/3} \log x), \tag{1.3}$$

which implies $\theta \leq 1/3$. In 2003, Huxley [4] proved the bound

$$\theta \leq 131/416. \tag{1.4}$$

In this paper, rather than pursuing further improvements of θ, we explore an alternative approach: an exact combinatorial decomposition of $D(x)$ via inclusion–exclusion on the number of distinct prime factors of n.

1.2 A new decomposition

For a positive integer n, let $\omega(n)$ denote the number of distinct prime factors of n. For each integer $j \geq 2$, define

$$T_j(x) := \sum_{n \leq x} d(n) \binom{\omega(n)}{j} \tag{1.5}$$

where $\binom{\omega(n)}{j}$ denotes the usual binomial coefficient. The main identity of this paper is the following.

Theorem 1.1. For every real number $x \geq 1$,

$$D(x) = 1 + \sum_{i \geq 1} (i+1)\pi(x^{1/i}) + \sum_{j \geq 2} (-1)^j (j-1) T_j(x) \tag{1.6}$$

where $\pi(x)$ denotes the prime-counting function. Both sums have only finitely many nonzero terms for each fixed x: we have $\pi(x^{1/i}) = 0$ for all sufficiently large i, and $T_j(x) = 0$ once j exceeds the maximum possible value of $\omega(n)$ among integers $n \leq x$.

The three terms on the right-hand side of (1.6) correspond, respectively, to $n = 1$, prime powers $\omega(n) = 1$, and integers with $\omega(n) \geq 2$. The proof of Theorem 1.1 is given in Section 2.

Dirichlet's estimate gives $D(x) \sim x \log x$. By the prime number theorem [1], the prime-power contribution satisfies

$$\sum_{i \geq 1} (i+1)\pi(x^{1/i}) \sim \frac{2x}{\log x}. \tag{1.7}$$

This is $o(x \log x)$. Thus the first two terms in the identity are negligible on the scale $x \log x$. The main contribution must therefore come from the alternating sum

$$\sum_{j \geq 2} (-1)^j (j-1) T_j(x) \tag{1.8}$$

after cancellation. As the asymptotics below show, the individual terms $T_j(x)$ are large, and substantial cancellation is needed in the full alternating sum. The fixed-j asymptotics proved below give information about individual terms, but they do not by themselves determine the full alternating sum.

Example 1.2. At $x = 10000$, direct computation gives

$$T_2(10000) = 297{,}078 \qquad T_3(10000) = 140{,}640$$
$$T_4(10000) = 26{,}674 \qquad T_5(10000) = 1{,}216$$

and $T_j(10000) = 0$ for $j \geq 6$. The prime-power contribution equals 2,711, and the identity (1.6) yields

$$1 + 2{,}711 + (T_2 - 2T_3 + 3T_4 - 4T_5) = 1 + 2{,}711 + 90{,}956 = 93{,}668 = D(10000),$$

illustrating Theorem 1.1. Moreover, the ratio of $T_j(x)$ to the leading term in Theorem 1.3 converges to 1 only very slowly: at $x = 10^9$, this ratio is 0.43 for $j = 2$ and 0.21 for $j = 3$. This slow convergence motivates the refined asymptotic expansion of $T_j(x)$ developed in Section 6.

1.3 Main results

To analyze $T_j(x)$, we consider the two-variable Dirichlet series

$$F(s, y) := \sum_{n \geq 1} \frac{d(n)(1+y)^{\omega(n)}}{n^s}.$$

Here and below, the definition of $T_j(x)$ is extended to all $j \geq 0$ by the same binomial-coefficient formula. The factor $(1+y)^{\omega(n)}$ is chosen so that its expansion in y produces the coefficients $\binom{\omega(n)}{j}$; in Section 4, the series $F(s, y)$ will be used to study the corresponding generating partial sums. The series $F(s, y)$ admits the factorization

$$F(s, y) = \zeta(s)^{2+2y} H(s, y), \tag{1.9}$$

where, for y in a small disk around 0, the function $H(s, y)$ is holomorphic and uniformly bounded in a strip $\mathrm{Re}(s) > 1 - \delta$. This factorization isolates the singularity of $F(s, y)$ at $s = 1$, and it is the starting point of the Selberg–Delange method [5, Chapter II.5].

Our main asymptotic result is the following.

Theorem 1.3. For each fixed integer $j \geq 2$, as $x \to \infty$,

$$\begin{aligned} T_j(x) = {} & \frac{2^j}{j!}\, x \log x \,(\log\log x)^j \\ & + \frac{2^{j-1}}{(j-1)!}\big(2(B_1 - 1) - P(2)\big) x \log x \,(\log\log x)^{j-1} \\ & + O_j\big(x \log x \,(\log\log x)^{j-2}\big), \end{aligned} \tag{1.10}$$

where B_1 is the Meissel–Mertens constant and $P(2) = \sum_p p^{-2}$ is the value of the prime zeta function at 2. In particular,

$$T_j(x) \sim \frac{2^j}{j!}\, x \log x \,(\log\log x)^j. \tag{1.11}$$

The proof of Theorem 1.3 is given in Sections 4–6. The essential ingredients are the truncated Perron formula, contour deformation around $s = 1$, and a local Hankel analysis in the w-plane obtained via the substitution $w = (s-1)\log x$.

Theorem 1.3 is a fixed-j result. It describes the individual components $T_j(x)$ in the decomposition, but it does not by itself determine the full alternating sum in (1.6). For further discussion of this limitation, see Section 7.

1.4 Outline of the paper

The paper is organized as follows. In Section 2, we prove the identity in Theorem 1.1 by partitioning $D(x)$ according to $\omega(n)$ and applying an inclusion–exclusion argument to the part with $\omega(n) \geq 2$. In Section 3, we introduce the two-variable Dirichlet series $F(s, y)$ and establish its factorization $F(s, y) = \zeta(s)^{2+2y} H(s, y)$, together with the analytic properties of $H(s, y)$ needed in the sequel.

Sections 4–6 are devoted to the proof of Theorem 1.3. In Section 4, we apply the truncated Perron formula for $F(s, y)$ to represent the generating partial sum $A(x, y) := \sum_{n \leq x} d(n)(1 + y)^{\omega(n)}$ and deform the contour around $s = 1$. In Section 5, we carry out the local Hankel analysis via the substitution $w = (s - 1)\log x$ and evaluate the main and lower-order contributions. In Section 6, we extract the asymptotic formula for $T_j(x)$ from that of $A(x, y)$ by taking the j-th derivative at $y = 0$, thereby completing the proof of Theorem 1.3.

Finally, in Section 7, we explain why the fixed-j asymptotics do not determine the full alternating sum in the decomposition of $D(x)$, and indicate the need for estimates uniform in j or for a direct analysis near $y = -1$.

2 Proof of the identity

In this section, we prove the identity stated in Theorem 1.1. The proof proceeds by partitioning the sum $D(x) = \sum_{n \leq x} d(n)$ according to the value of $\omega(n)$, and then applying an inclusion–exclusion argument to the composite part.

Proof of Theorem 1.1. For fixed x, all sums below are finite. Indeed, $\pi(x^{1/i}) = 0$ for all sufficiently large i, and $\binom{\omega(n)}{j} = 0$ whenever $j > \omega(n)$.

Partition the integers n with $1 \leq n \leq x$ into three disjoint classes:

(i) $n = 1$ (equivalently, $\omega(n) = 0$);

(ii) prime powers, i.e., $\omega(n) = 1$;

(iii) composite integers with $\omega(n) \geq 2$.

Writing $D(x)$ as the sum of contributions from each class, we have

$$D(x) = d(1) + \sum_{\substack{n \leq x \\ \omega(n)=1}} d(n) + \sum_{\substack{n \leq x \\ \omega(n)\geq 2}} d(n). \tag{2.1}$$

Since $d(1) = 1$, the first term equals 1.

For the second term, every integer n with $\omega(n) = 1$ is of the form $n = p^i$ for a unique prime p and a unique integer $i \geq 1$, and $d(p^i) = i + 1$. The number of prime powers $p^i \leq x$ with exponent exactly i equals $\pi(x^{1/i})$. Hence

$$\sum_{\substack{n \leq x \\ \omega(n)=1}} d(n) = \sum_{i \geq 1} (i + 1)\pi(x^{1/i}). \tag{2.2}$$

For the third term, we apply inclusion–exclusion on the set of distinct prime divisors of n. For each integer $j \geq 2$, recall the definition

$$T_j(x) = \sum_{n \leq x} d(n) \binom{\omega(n)}{j}.$$

We claim that

$$\sum_{\substack{n\le x\\ \omega(n)\ge 2}} d(n) = \sum_{j\ge 2}(-1)^j (j-1)T_j(x). \tag{2.3}$$

To prove (2.3), interchange the order of summation on the right-hand side. This involves only finite sums, so no convergence issue occurs:

$$\sum_{j\ge 2}(-1)^j (j-1)T_j(x) = \sum_{n\le x} d(n) \sum_{j\ge 2}(-1)^j (j-1)\binom{\omega(n)}{j}.$$

For n with $\omega(n)\in\{0,1\}$, we have $\binom{\omega(n)}{j}=0$ for all $j\ge 2$, so such n do not contribute. For n with $\omega(n)=m\ge 2$, we use the following combinatorial identity.

Lemma 2.1. For every integer $m\ge 2$,

$$\sum_{j=2}^{m}(-1)^j (j-1)\binom{m}{j} = 1. \tag{2.4}$$

Proof of the lemma. Starting from the standard identities

$$\sum_{j=0}^{m}(-1)^j \binom{m}{j} = 0 \quad \text{and} \quad \sum_{j=0}^{m}(-1)^j\, j\binom{m}{j} = \frac{d}{dx}(1-x)^m\bigg|_{x=1} = 0 \text{ (for } m\ge 2),$$

we combine them to obtain

$$\sum_{j=0}^{m}(-1)^j (j-1)\binom{m}{j} = 0 \quad (\text{for } m\ge 2).$$

Separating the $j=0$ and $j=1$ terms:

$$1\cdot(-1) + 0\cdot\binom{m}{1} + \sum_{j=2}^{m}(-1)^j (j-1)\binom{m}{j} = 0,$$

which yields

$$\sum_{j=2}^{m}(-1)^j (j-1)\binom{m}{j} = 1. \qquad \square$$

Applying the lemma, we obtain

$$\sum_{n\le x} d(n)\sum_{j\ge 2}(-1)^j (j-1)\binom{\omega(n)}{j} = \sum_{\substack{n\le x\\ \omega(n)\ge 2}} d(n)\cdot 1,$$

which establishes (2.3).

Combining the contributions from all three classes yields

$$D(x) = 1 + \sum_{i\ge 1}(i+1)\pi(x^{1/i}) + \sum_{j\ge 2}(-1)^j (j-1)T_j(x),$$

completing the proof. $\square$

3 Dirichlet series and factorization

In this section, we introduce the two-variable Dirichlet series $F(s,y)$ and establish its key analytic properties. The main output is the factorization $F(s,y) = \zeta(s)^{2+2y} H(s,y)$, where $H(s,y)$ is holomorphic in a neighborhood of $s=1$ and uniformly bounded in a small disk around $y=0$.

3.1 Euler product

Recall the definition

$$F(s,y) := \sum_{n\geq 1} \frac{d(n)(1+y)^{\omega(n)}}{n^s}. \tag{3.1}$$

Since the arithmetic function $n \mapsto d(n)(1+y)^{\omega(n)}$ is multiplicative in n for each fixed y, the series admits an Euler product representation for $\mathrm{Re}(s) > 1$ [1].

Lemma 3.1. For $\mathrm{Re}(s) > 1$ and every $y \in \mathbb{C}$,

$$F(s,y) = \prod_p \left(1 + (1+y)\left((1-p^{-s})^{-2} - 1\right)\right), \tag{3.2}$$

where the product is over all primes p.

Proof. By multiplicativity, for $\mathrm{Re}(s) > 1$ we have

$$F(s,y) = \sum_{n\geq 1} \frac{d(n)(1+y)^{\omega(n)}}{n^s} = \prod_p \left(\sum_{k\geq 0} \frac{d(p^k)(1+y)^{\omega(p^k)}}{p^{ks}} \right).$$

For $k=0$, the summand equals 1. For $k \geq 1$, we have $d(p^k) = k+1$ and $\omega(p^k) = 1$, so the local factor becomes

$$1 + (1+y)\sum_{k\geq 1} \frac{k+1}{p^{ks}}.$$

Setting $u := p^{-s}$ and using the standard identity

$$\sum_{k\geq 1} (k+1)u^k = \frac{1}{(1-u)^2} - 1 \quad \text{for } |u| < 1,$$

the local factor equals $1 + (1+y)\big((1-p^{-s})^{-2} - 1\big)$, which proves (3.2). □

3.2 Factorization isolating the singularity

The Riemann zeta function admits the Euler product [1] $\zeta(s) = \prod_p (1-p^{-s})^{-1}$ for $\mathrm{Re}(s) > 1$, so that $\zeta(s)^{2+2y} = \prod_p (1-p^{-s})^{-(2+2y)}$ on the same half-plane, provided a branch of the logarithm is fixed. We use the principal branch, which is analytic in s in a neighborhood of any point with $\mathrm{Re}(s) > 1$ and in y in a neighborhood of 0.

Define

$$H(s,y) := \prod_p (1-p^{-s})^{2+2y} \left(1 + (1+y)\left((1-p^{-s})^{-2} - 1\right)\right). \tag{3.3}$$

Theorem 3.2. For $\mathrm{Re}(s) > 1$ and y in a neighborhood of 0,

$$F(s,y) = \zeta(s)^{2+2y} H(s,y). \tag{3.4}$$

Proof. This follows by combining Lemma 3.1 with the Euler product for $\zeta(s)^{2+2y}$ and regrouping factors. $\square$

The purpose of the factorization (3.4) is to isolate the singular behavior of $F(s,y)$ at $s=1$ into the factor $\zeta(s)^{2+2y}$, leaving $H(s,y)$ as a well-behaved function in a larger region, as we now establish.

3.3 Analytic properties of $H(s,y)$

We show that the Euler product (3.3) defines a holomorphic function on a strip containing $\mathrm{Re}(s)=1$, uniformly for y in a small disk around 0. The key is that each local factor $H_p(s,y)$ deviates from 1 by a quantity of order $p^{-2\,\mathrm{Re}(s)}$.

Lemma 3.3. Fix $\eta \in (0,1/2)$ and $\delta \in (0,1/2)$. Write $u := p^{-s}$ and denote the local factor by

$$H_p(s,y) := (1-u)^{2+2y}\left(1+(1+y)\left((1-u)^{-2}-1\right)\right).$$

Then for $|y| \le \eta$ and $\mathrm{Re}(s) \ge 1-\delta$,

$$H_p(s,y) = 1 + O_{\eta,\delta}(u^2), \tag{3.5}$$

where the implied constant depends only on η and δ.

Proof. The condition $\mathrm{Re}(s) \ge 1-\delta$ implies that, for every prime $p \ge 2$,

$$|u| = p^{-\mathrm{Re}(s)} \le p^{-(1-\delta)} \le 2^{-(1-\delta)} < 1,$$

since $\delta < 1/2$. In particular, $|u|$ is bounded away from 1, and the power series expansions

$$(1-u)^{-2}-1 = 2u+3u^2+4u^3+\cdots = 2u+O_\delta(u^2),$$

$$(1-u)^{2+2y} = 1-(2+2y)u+\binom{2+2y}{2}u^2+\cdots = 1-(2+2y)u+O_{\eta,\delta}(u^2),$$

converge absolutely, with remainders uniform for $|y| \le \eta$ and $\mathrm{Re}(s) \ge 1-\delta$. Multiplying the expansions:

$$H_p(s,y) = \left(1-(2+2y)u+O_{\eta,\delta}(u^2)\right)\left(1+(1+y)(2u+O_\delta(u^2))\right).$$

The coefficient of u^1 on the right is $-(2+2y)+2(1+y)=0$. Hence $H_p(s,y)=1+O_{\eta,\delta}(u^2)$, proving (3.5). $\square$

Proposition 3.4. Fix $\eta \in (0,1/2)$ and $\delta \in (0,1/2)$, and let

$$\mathcal{R}_{\delta,\eta} := \{(s,y) \in \mathbb{C}^2 : \mathrm{Re}(s) \ge 1-\delta,\ |y| \le \eta\}. \tag{3.6}$$

Then the partial products of the Euler product (3.3) converge absolutely and uniformly on $\mathcal{R}_{\delta,\eta}$ to a function $H(s,y)$. For every y with $|y| \le \eta$, the function $H(s,y)$ is holomorphic in s on the open strip $\mathrm{Re}(s) > 1-\delta$. Moreover, $H(s,y)$ is jointly holomorphic in (s,y) on $\mathrm{int}(\mathcal{R}_{\delta,\eta})$, and H is uniformly bounded on $\mathcal{R}_{\delta,\eta}$.

Proof. By Lemma 3.3, for every $(s, y) \in \mathcal{R}_{\delta,\eta}$ and every prime p,

$$|H_p(s, y) - 1| \leq C_{\eta,\delta}\, p^{-2(1-\delta)}, \tag{3.7}$$

where the constant $C_{\eta,\delta}$ depends only on η and δ. Since $\delta < 1/2$, we have $2(1 - \delta) > 1$, and the series $\sum_n n^{-2(1-\delta)}$ converges; in particular, $\sum_p p^{-2(1-\delta)}$ converges. By the Weierstrass M-test, the series

$$\sum_p |H_p(s, y) - 1|$$

converges uniformly on $\mathcal{R}_{\delta,\eta}$. Hence the infinite product criterion applies: if $\sum_n |u_n|$ converges uniformly, then the product $\prod_n (1 + u_n)$ converges absolutely and uniformly. Therefore the partial products $\prod_{p\leq P} H_p(s, y)$ converge absolutely and uniformly on $\mathcal{R}_{\delta,\eta}$ to a limit, which we denote $H(s, y)$.

To deduce holomorphy in s, fix $y_0 \in \mathbb{C}$ with $|y_0| \leq \eta$ and let $s_0 \in \mathbb{C}$ be any point with $\mathrm{Re}(s_0) > 1 - \delta$. Choose $r > 0$ small enough that $r < \mathrm{Re}(s_0) - (1 - \delta)$, and define

$$K := \overline{D}(s_0, r) = \{s \in \mathbb{C} : |s - s_0| \leq r\}. \tag{3.8}$$

The set K is closed and bounded in $\mathbb{C}$, so by the Heine–Borel theorem it is compact. By the choice of r, every $s \in K$ satisfies

$$\mathrm{Re}(s) \geq \mathrm{Re}(s_0) - r > 1 - \delta,$$

and hence $K \times \{y_0\} \subset \mathcal{R}_{\delta,\eta}$. The uniform convergence of the partial products on $\mathcal{R}_{\delta,\eta}$ established above therefore restricts to uniform convergence of $\prod_{p\leq P} H_p(s, y_0)$ on K as $P \to \infty$.

Each finite partial product $\prod_{p\leq P} H_p(s, y_0)$ is holomorphic in s on the open disk $D(s_0, r) = \mathrm{int}(K)$, since each factor $H_p(\cdot, y_0)$ is. By the theorem on uniform limits of holomorphic functions (Weierstrass), the limit $H(\cdot, y_0)$ is holomorphic on $D(s_0, r)$, and in particular at s_0. Since s_0 with $\mathrm{Re}(s_0) > 1 - \delta$ was arbitrary, $H(\cdot, y_0)$ is holomorphic on the open strip $\mathrm{Re}(s) > 1 - \delta$. Since y_0 with $|y_0| \leq \eta$ was also arbitrary, the holomorphy claim in s is proved.

For joint holomorphy, each finite partial product $\prod_{p\leq P} H_p(s, y)$ is holomorphic in (s, y) on $\mathrm{int}(\mathcal{R}_{\delta,\eta})$, since each factor $H_p(s, y)$ is. The uniform convergence of the partial products on $\mathcal{R}_{\delta,\eta}$ established above restricts to uniform convergence on every compact subset of $\mathrm{int}(\mathcal{R}_{\delta,\eta})$, since $\mathrm{int}(\mathcal{R}_{\delta,\eta}) \subset \mathcal{R}_{\delta,\eta}$. By the theorem on uniform limits of holomorphic functions in two complex variables, the limit $H(s, y)$ is jointly holomorphic on $\mathrm{int}(\mathcal{R}_{\delta,\eta})$.

Uniform boundedness on $\mathcal{R}_{\delta,\eta}$ follows from (3.7):

$$|H(s, y)| \leq \prod_p \big(1 + |H_p(s, y) - 1|\big) \leq \exp\left(\sum_p |H_p(s, y) - 1|\right) \leq \exp\left(C_{\eta,\delta} \sum_p p^{-2(1-\delta)}\right),$$

the right-hand side being finite and independent of $(s, y) \in \mathcal{R}_{\delta,\eta}$. ☐

Lemma 3.5. The function $H(s, y)$ satisfies

$$H(s, 0) \equiv 1 \quad \text{on } \mathrm{Re}(s) > 1/2. \tag{3.9}$$

In particular, $H(1, 0) = 1$.

Proof. On $\mathrm{Re}(s) > 1$, the factorization (3.4) specialized at $y = 0$ reads

$$F(s, 0) = \zeta(s)^2\, H(s, 0).$$

On the other hand, the Dirichlet series defining F gives

$$F(s,0) = \sum_{n\geq 1} \frac{d(n)}{n^s} = \zeta(s)^2 \quad \text{on } \operatorname{Re}(s) > 1.$$

Comparing the two expressions yields $\zeta(s)^2 H(s,0) = \zeta(s)^2$ on $\operatorname{Re}(s) > 1$. Since $\zeta(s) \neq 0$ on $\operatorname{Re}(s) > 1$ (which follows directly from the Euler product representation of ζ), we may divide both sides by $\zeta(s)^2$ to obtain

$$H(s,0) \equiv 1 \quad \text{on } \operatorname{Re}(s) > 1.$$

By Proposition 3.4, applied with any $\delta \in (0,1/2)$, the function $H(s,0)$ is holomorphic on $\operatorname{Re}(s) > 1-\delta$; letting $\delta \to 1/2$ shows that $H(s,0)$ is holomorphic on the connected open set $\operatorname{Re}(s) > 1/2$. The constant function 1 is also holomorphic there, and the two functions agree on the open subset $\operatorname{Re}(s) > 1$. By the identity theorem, they agree on all of $\operatorname{Re}(s) > 1/2$. In particular, $H(1,0) = 1$. □

4 Perron formula and contour deformation

To extract the asymptotic behavior of $T_j(x)$ from the Dirichlet series $F(s,y)$, we work with the generating partial sum

$$A(x,y) := \sum_{n\leq x} d(n)(1+y)^{\omega(n)}, \tag{4.1}$$

which satisfies the generating relation

$$A(x,y) = \sum_{j\geq 0} T_j(x) y^j. \tag{4.2}$$

Throughout this section, we initially fix one parameter:

- $\eta \in (0,1/2)$, controlling the size of the disk $|y| \leq \eta$.

After the contour depth δ' is fixed, we may shrink η if necessary for the contour estimates.

The fixed strip depth δ' and the truncation-dependent quantity λ_T will be chosen together in Section 4.3. This keeps the Euler factor estimates, the zero-free region, and the contour growth bound tied to the same strip parameter.

We also abbreviate $\alpha := 2+2y$. Thus the factorization (3.4) becomes $F(s,y) = \zeta(s)^\alpha H(s,y)$. For $|y| \leq \eta$ with $\eta < 1/2$, the exponent α ranges in a disk of radius $2\eta < 1$ around 2, so in particular α is bounded away from non-positive integers.

4.1 The truncated Perron formula

We quote the following standard form of the truncated Perron formula (see, e.g., []).

Theorem 4.1 (Truncated Perron formula)**.** Let $f(s) = \sum_{n\geq 1} a_n n^{-s}$ be a Dirichlet series with abscissa of absolute convergence σ_a. For real numbers $c > \sigma_a$, $x \geq 2$, and $T \geq 1$ with $x \notin \mathbb{Z}$,

$$\sum_{n\leq x} a_n = \frac{1}{2\pi i}\int_{c-iT}^{c+iT} f(s)\frac{x^s}{s}\,ds + R(x,T), \tag{4.3}$$

where the truncation error satisfies

$$R(x,T) \ll x^c \sum_{n\geq 1} \frac{|a_n|}{n^c} \frac{1}{1+T\left|\log \frac{x}{n}\right|}. \tag{4.4}$$

If $x = N$ is an integer, we do not apply the Perron formula directly at N. Instead, we apply it at

$$X := N + \frac{1}{2}.$$

Then $X \notin \mathbb{Z}$ and

$$A(N, y) = A(X, y),$$

because the integers satisfying $n \le N$ are exactly the integers satisfying $n \le N + 1/2$. Moreover,

$$X = N + O(1), \qquad \log X = \log N + O(1/N),$$

so every asymptotic estimate obtained for X transfers back to N. Thus it is enough in the Perron argument to work with non-integral x; the integer case follows from this replacement.

We apply Theorem 4.1 to the Dirichlet series $F(s, y) = \sum_{n\ge1} d(n)(1 + y)^{\omega(n)}n^{-s}$. For $|y| \le \eta$ with η sufficiently small, the series converges absolutely for $\mathrm{Re}(s) > 1$, so we may take any $c > 1$. Choose, for instance, $c = 1 + 1/\log x$, which is the standard choice to balance the main term size and the truncation error.

For $x \ge 2$ with $x \notin \mathbb{Z}$, $|y| \le \eta$, and $T \ge 1$, put $c = 1 + 1/\log x$. Theorem 4.1 gives

$$A(x, y) = \frac{1}{2\pi i} \int_{c-iT}^{c+iT} F(s, y) \frac{x^s}{s} \, ds + E_{\mathrm{Perron}}(x, T, y), \tag{4.5}$$

where

$$E_{\mathrm{Perron}}(x, T, y) \ll_\eta x^c \sum_{n\ge1} \frac{d(n)(1 + \eta)^{\omega(n)}}{n^c} \frac{1}{1 + T \left|\log \frac{x}{n}\right|}. \tag{4.6}$$

Indeed, applying (4.4) with $a_n = d(n)(1+y)^{\omega(n)}$ and using $|1+y| \le 1+\eta$ gives the displayed bound.

4.2 Analytic preliminaries

To control the contour integrals arising from the deformation in Section 4.3, we will need:

(a) a region in which $\zeta(s)$ does not vanish, so that the contour can avoid its zeros;

(b) a continuous branch of $\zeta(s)^{2+2y}$ on a neighborhood of the contour;

(c) a uniform bound on $|\zeta(s)|$ along vertical lines in the critical strip;

(d) a corresponding bound on $\arg \zeta(s)$ on the chosen branch.

We collect these analytic ingredients here, before constructing the contour itself in Section 4.3.

Zero-free region

For the contour deformation to be valid, we need a region in which $\zeta(s) \ne 0$. We use the following classical result of de la Vallée Poussin, quoted without proof; see, e.g., [].

Theorem 4.2 (de la Vallée Poussin zero-free region)**.** There exists an absolute constant $c_0 \in (0, 1/2)$ such that $\zeta(s) \ne 0$ on the open region

$$\mathcal{Z} := \left\{ s = \sigma + it \in \mathbb{C} : \sigma > 1 - \frac{c_0}{\log(|t| + 2)} \right\} \setminus \{1\}. \tag{4.7}$$

The boundary of $\mathcal{Z}$ is curved, and integrating along it would complicate the estimation of the resulting contour pieces. We will instead use a fixed vertical line lying strictly inside $\mathcal{Z}$ for all $|t| \le T$; the precise depth of this line is fixed in Section 4.3.

Continuous branch of $\zeta(s)^\alpha$

To make sense of $\zeta(s)^\alpha = \zeta(s)^{2+2y}$ on a neighborhood of the contour, we specify a branch on the region

$$\mathcal{D} := \mathcal{Z} \setminus \mathcal{S}, \tag{4.8}$$

where $\mathcal{S} := \{s \in \mathbb{C} : \Im(s) = 0,\ \mathrm{Re}(s) \leq 1\}$ is the slit along the real axis to the left of $s = 1$. The slit $\mathcal{S}$ disconnects the segment $\mathcal{Z} \cap \mathcal{S}$ from the rest of $\mathcal{Z}$ from above and below, rendering $\mathcal{D}$ simply connected.[1]

On $\mathcal{D}$, both $\zeta(s)$ and $\zeta'(s)$ are holomorphic, and $\zeta(s) \neq 0$. Fix any real number $c_* > 1$ (for example $c_* = 2$); since $\zeta(c_*) > 0$, the principal logarithm $\log \zeta(c_*) \in \mathbb{R}$ is well-defined. Define

$$L(s) := \log \zeta(c_*) + \int_{c_*}^{s} \frac{\zeta'(\omega)}{\zeta(\omega)}\, d\omega \qquad (s \in \mathcal{D}), \tag{4.9}$$

where the integral is taken along any path in $\mathcal{D}$ from c_* to s. Since $\mathcal{D}$ is simply connected and ζ'/ζ is holomorphic on $\mathcal{D}$, the integral is path-independent, and L is a well-defined holomorphic function on $\mathcal{D}$.

We claim $e^{L(s)} = \zeta(s)$ on $\mathcal{D}$. Indeed, $L'(s) = \zeta'(s)/\zeta(s)$ on $\mathcal{D}$, so

$$\frac{d}{ds}\big(e^{-L(s)}\zeta(s)\big) = e^{-L(s)}\big(\zeta'(s) - L'(s)\zeta(s)\big) = 0.$$

Hence $e^{-L(s)}\zeta(s)$ is constant on $\mathcal{D}$, and evaluating at $s = c_*$ gives $e^{-L(c_*)}\zeta(c_*) = e^{-\log\zeta(c_*)}\zeta(c_*) = 1$. Therefore $e^{L(s)} = \zeta(s)$ throughout $\mathcal{D}$, so L serves as a continuous branch of $\log \zeta$ on $\mathcal{D}$.

We define

$$\zeta(s)^\alpha := e^{\alpha L(s)} \qquad (s \in \mathcal{D},\ \alpha \in \mathbb{C}). \tag{4.10}$$

This is single-valued and holomorphic in s on $\mathcal{D}$, and entire in α. At $\alpha = 2$, the definition reduces to $\zeta(s)^2$ in the usual sense.

Vertical growth of $\zeta(s)$

We use the following uniform bound on $|\zeta(s)|$, obtained by combining the convexity bound with the trivial estimate $\zeta(\sigma + it) \ll \log|t|$ for $\sigma \geq 1$ (see, e.g., [] for the convexity bound).

Theorem 4.3 (Uniform bound on $\zeta(s)$ in a vertical strip)**.** Fix $\delta \in (0, 1/2)$. For every $\varepsilon > 0$,

$$|\zeta(\sigma + it)| \ll_{\delta,\varepsilon} |t|^{\delta/2+\varepsilon} \quad \text{for } 1 - \delta \leq \sigma \leq 1 + \delta \text{ and } |t| \geq 2, \tag{4.11}$$

where the implied constant is independent of t.

Sketch. For $1 - \delta \leq \sigma \leq 1$, the convexity bound gives $|\zeta(\sigma + it)| \ll_\varepsilon |t|^{(1-\sigma)/2+\varepsilon}$. Taking the supremum of $(1 - \sigma)/2$ over $\sigma \in [1 - \delta, 1]$ yields $\delta/2$, so $|\zeta(\sigma + it)| \ll_\varepsilon |t|^{\delta/2+\varepsilon}$ on this subrange.

For $1 \leq \sigma \leq 1 + \delta$, the standard estimate $|\zeta(\sigma + it)| \ll \log|t|$ holds, and $\log|t| \ll_\varepsilon |t|^\varepsilon \leq |t|^{\delta/2+\varepsilon}$. Combining the two ranges gives (4.11). □

[1] Removing the slit from $s = 1$ to the boundary cuts the only possible loop around the puncture $s = 1$, just as $\mathbb{C} \setminus (-\infty, 0]$ is the standard simply connected branch domain for the logarithm. This is the topological input needed for path independence in the definition of $L(s)$.

Bound on $\arg\zeta$

We also use the following standard estimate on the branch determined by (4.9); see, e.g., [].

Theorem 4.4 (Bound on $\arg\zeta$)**.** There exists an absolute constant $C_{\arg} > 0$ such that, for every $s = \sigma + it \in \mathcal{D}$ with $|t| \geq 2$,

$$|\arg\zeta(s)| \leq C_{\arg}\log|t|, \tag{4.12}$$

where the branch of $\arg\zeta$ is the one determined by the logarithm L on $\mathcal{D}$.

4.3 Contour decomposition

We now introduce the parameters governing the contour deformation and construct the deformed contour.

Depth of the deformation

Fix once and for all an absolute constant

$$\delta' \in \left(\frac{c_0}{2\log 3}, \frac{c_0}{2}\right), \tag{4.13}$$

where c_0 is the constant of Theorem 4.2; the interval is non-empty since $\log 3 > 1$. This range of δ' is chosen so that the deformed contour remains in the fixed strip used in Section 4.4.

For each truncation height $T \geq 1$, define

$$\lambda_T := \frac{c_0}{2\log(T+2)}. \tag{4.14}$$

By construction, $\lambda_T \to 0$ as $T \to \infty$, and for every $T \geq 1$,

$$\lambda_T \leq \frac{c_0}{2\log 3} < \delta', \tag{4.15}$$

so the depth $\sigma_0(T) := 1 - \lambda_T$ of the deformed contour satisfies $\sigma_0(T) > 1 - \delta'$. Consequently, the contour $\mathcal{C}(T,\rho)$ defined below lies in $\mathcal{R}_{\delta',\eta}$ for every $T \geq 1$ and every $|y| \leq \eta$. In particular, Proposition 3.4 applies on the contour with $\delta = \delta'$.

Finally, the choice $c = 1 + 1/\log x$ in the Perron representation (4.5) satisfies $c \leq 1 + \delta'$ for all x sufficiently large, ensuring that the abscissa c lies inside the strip $1 - \delta' \leq \sigma \leq 1 + \delta'$ used in the contour estimates. We assume this hereafter.

The deformed contour

Fix a truncation height $T \geq 1$, and set

$$\sigma_0 := \sigma_0(T) := 1 - \lambda_T. \tag{4.16}$$

Choose a parameter $\rho > 0$ with $\rho < \lambda_T$; its precise choice as a function of x will be made in Section 5. The notation $i0^{\pm}$ below denotes boundary values: the two Hankel arms are first taken at heights $\pm\varepsilon$ with $\varepsilon > 0$, and the limit $\varepsilon \to 0^+$ is taken after the corresponding integrals are formed. Thus the approximating contours lie in the slit domain $\mathcal{D}$ before the boundary value limit is taken.

We deform the original Perron segment $[c - iT, c + iT]$ leftward, around the singularity at $s = 1$, into a contour $\mathcal{C} = \mathcal{C}(T,\rho)$ traversed from $c - iT$ to $c + iT$. It consists of the following five pieces:

(1) $\mathcal{C}_{\rm bot}$: the horizontal segment from $c - iT$ to $\sigma_0 - iT$.

(2) $\mathcal{C}^-_{\rm left}$: the vertical segment from $\sigma_0 - iT$ to $\sigma_0 + i0^-$.

(3) $\mathcal{H}_{\rm loc}$: a local Hankel contour around $s = 1$, consisting of

  (3a) the lower arm from $\sigma_0 + i0^-$ to $1 - \rho + i0^-$;

  (3b) the circle $|s - 1| = \rho$, traversed from $1-\rho+i0^-$ to $1-\rho+i0^+$ counterclockwise through the right side of $s = 1$;

  (3c) the upper arm from $1 - \rho + i0^+$ to $\sigma_0 + i0^+$.

(4) $\mathcal{C}^+_{\rm left}$: the vertical segment from $\sigma_0 + i0^+$ to $\sigma_0 + iT$.

(5) $\mathcal{C}_{\rm top}$: the horizontal segment from $\sigma_0 + iT$ to $c + iT$.

The five pieces are joined consecutively, so $\mathcal{C}$ has the same initial and terminal points as the original Perron contour. The Hankel arms are understood as boundary values along the slit $\mathcal{S}$; in the local representation near $s = 1$, their discontinuity is carried by the factor $(s-1)^{-2-2y}$ and produces the local contribution I_H.

Verification that the contour avoids ζ-zeros

We claim that $\mathcal{C} \setminus \{1\} \subset \mathcal{Z}$, where $\mathcal{Z}$ is the zero-free region of Theorem 4.2. For the horizontal pieces $\mathcal{C}_{\rm bot}$ and $\mathcal{C}_{\rm top}$,

$$\mathrm{Re}(s) \geq \sigma_0(T) = 1 - \frac{c_0}{2\log(T+2)} > 1 - \frac{c_0}{\log(|\Im s| + 2)}. \tag{4.17}$$

For the vertical pieces $\mathcal{C}^{\pm}_{\rm left}$,

$$\mathrm{Re}(s) = \sigma_0(T) = 1 - \frac{c_0}{2\log(T+2)} > 1 - \frac{c_0}{\log(|\Im s| + 2)}. \tag{4.18}$$

For the local Hankel contour $\mathcal{H}_{\rm loc}$, the two horizontal arms (3a) and (3c) satisfy $\mathrm{Re}(s) \geq \sigma_0(T)$ and $|\Im s| \leq \rho$, hence are covered by the same estimate as the vertical pieces. For the small circle (3b),

$$\mathrm{Re}(s) \geq 1 - \rho > 1 - \lambda_T = 1 - \frac{c_0}{2\log(T+2)} > 1 - \frac{c_0}{\log(|\Im s| + 2)}. \tag{4.19}$$

Therefore $\mathcal{C} \setminus \{1\} \subset \mathcal{Z}$. Combined with the boundary-value convention for the Hankel arms, the approximating contours with heights $\pm\varepsilon$ are contained in $\mathcal{D}$ away from $s = 1$. The integrand $F(s, y)x^s/s$ is therefore well-defined and holomorphic on a neighborhood of each approximating contour, and the contour $\mathcal{C}$ is understood as the corresponding boundary-value limit.

Cauchy's theorem

For $\varepsilon > 0$, replace the two Hankel arms along the slit by parallel paths at heights $-\varepsilon$ and $+\varepsilon$, and denote the resulting deformed contour by $\mathcal{C}_\varepsilon$. The Perron contour and $\mathcal{C}_\varepsilon$ share the same endpoints $c \pm iT$. Together, the Perron contour traversed from $c - iT$ to $c + iT$ and $\mathcal{C}_\varepsilon$ traversed in the reverse direction form a closed loop Γ_ε, oriented counterclockwise. The keyhole formed by the approximating Hankel contour and the slit $\mathcal{S}$ excludes the point $s = 1$ from the interior of Γ_ε, so the interior of Γ_ε is contained in the rectangle

$$\mathcal{R}_\Gamma := \{s = \sigma + it : \sigma_0(T) \leq \sigma \leq c,\ |t| \leq T\} \setminus \{1\}, \tag{4.20}$$

with the slit further excluded. For every $s \in \mathcal{R}_\Gamma$,

$$\mathrm{Re}(s) \geq \sigma_0(T) = 1 - \frac{c_0}{2\log(T+2)} > 1 - \frac{c_0}{\log(|\Im s|+2)}, \tag{4.21}$$

so $\mathcal{R}_\Gamma \subset \mathcal{Z}$. Combined with the on-contour verifications (4.17)–(4.19), this shows that Γ_ε together with its interior, with the slit removed, is contained in the slit domain $\mathcal{D}$. Hence $F(s,y)x^s/s$ is holomorphic on an open neighborhood of Γ_ε and its interior. By Cauchy's theorem,

$$\oint_{\Gamma_\varepsilon} F(s,y)\frac{x^s}{s}\,ds = 0, \tag{4.22}$$

which yields

$$\int_{c-iT}^{c+iT} F(s,y)\frac{x^s}{s}\,ds = \int_{\mathcal{C}_\varepsilon} F(s,y)\frac{x^s}{s}\,ds.$$

Letting $\varepsilon \to 0^+$ gives the boundary-value contour $\mathcal{C}$ and hence

$$\int_{c-iT}^{c+iT} F(s,y)\frac{x^s}{s}\,ds = \int_{\mathcal{C}} F(s,y)\frac{x^s}{s}\,ds. \tag{4.23}$$

Decomposition of $A(x,y)$

For non-integral x, combining (4.23) with the Perron representation (4.5), we obtain

$$A(x,y) = I_H + I_{\mathrm{top}} + I_{\mathrm{bot}} + I_{\mathrm{left}} + E_{\mathrm{Perron}}(x,T,y), \tag{4.24}$$

where, for $* \in \{\mathrm{top},\ \mathrm{bot},\ \mathrm{left}\}$,

$$I_* := \frac{1}{2\pi i}\int_{\mathcal{C}_*} F(s,y)\frac{x^s}{s}\,ds, \tag{4.25}$$

with $\mathcal{C}_{\mathrm{left}} := \mathcal{C}_{\mathrm{left}}^- \cup \mathcal{C}_{\mathrm{left}}^+$, and

$$I_H := \frac{1}{2\pi i}\int_{\mathcal{H}_{\mathrm{loc}}} F(s,y)\frac{x^s}{s}\,ds. \tag{4.26}$$

Writing $\mathcal{H}_{\mathrm{loc}} = \mathcal{H}_{\mathrm{loc}}^- \cup \mathcal{H}_{\mathrm{loc}}^\circ \cup \mathcal{H}_{\mathrm{loc}}^+$ for the lower arm, the small circle, and the upper arm, respectively, fixes the geometric decomposition of the local contour. The lower arm is oriented from $\sigma_0 + i0^-$ to $1-\rho+i0^-$, the small circle is oriented counterclockwise from $1-\rho+i0^-$ to $1-\rho+i0^+$ through the right side of $s=1$, and the upper arm is oriented from $1-\rho+i0^+$ to σ_0+i0^+.

The local Hankel contribution I_H contains the main asymptotic behavior and will be evaluated in Section 5. The remaining contributions $I_{\mathrm{top}}, I_{\mathrm{bot}}, I_{\mathrm{left}}$ are estimated in Section 4.4 and shown to be of lower order than the main term, after a suitable choice of $T = T(x)$.

4.4 Estimates of the contour pieces

In this section, we estimate the three nonlocal contour contributions $I_{\mathrm{top}}, I_{\mathrm{bot}}, I_{\mathrm{left}}$ appearing in the decomposition (4.24), and combine them with the Perron truncation error E_{Perron} to obtain a single error bound. The local Hankel contribution I_H is deferred to Section 5.

We first collect the two bounds for $F(s,y)$ used below.

Lemma 4.5 (Bounds for $F(s,y)$ on the deformed contour)**.** Let $0 < \eta < 1/2$, and let δ' be fixed as in (4.13). Let $\varepsilon > 0$ and put $\alpha = 2 + 2y$. Define

$$B := B(\eta,\delta',\varepsilon) := (2+2\eta)\Big(\tfrac{\delta'}{2} + \varepsilon\Big) + 2\eta C_{\mathrm{arg}}. \tag{4.27}$$

Then the following estimates hold uniformly for $|y| \leq \eta$.

1. If $s = \sigma + it \in \mathcal{D}$, $|t| \geq 2$, and $1 - \delta' \leq \sigma \leq 1 + \delta'$, then
$$|F(s, y)| \ll_{\eta,\delta',\varepsilon} |t|^B. \tag{4.28}$$

2. If $T \geq 1$, $s = 1 - \lambda_T + it$, and $|t| \leq 2$, then
$$|F(s, y)| \ll_{\eta,\delta'} \lambda_T^{-2-2\eta} \ll_{\eta,\delta'} (\log(T + 2))^{2+2\eta}. \tag{4.29}$$

Proof. For the first estimate, let $s = \sigma + it \in \mathcal{D}$ with $|t| \geq 2$ and $1 - \delta' \leq \sigma \leq 1 + \delta'$. On $\mathcal{D}$, the chosen branch gives
$$F(s, y) = \zeta(s)^\alpha H(s, y), \qquad \zeta(s)^\alpha = \exp(\alpha L(s)).$$
Writing $L(s) = \log |\zeta(s)| + i \arg \zeta(s)$ on this branch,
$$|\zeta(s)^\alpha| = |\zeta(s)|^{\mathrm{Re}(\alpha)} \exp\big(-\Im(\alpha) \arg \zeta(s)\big).$$
Since $|y| \leq \eta$, we have $\mathrm{Re}(\alpha) \leq 2 + 2\eta$ and $|\Im(\alpha)| \leq 2\eta$. Hence
$$|F(s, y)| \leq |H(s, y)|(1 + |\zeta(s)|)^{2+2\eta} \exp\big(2\eta| \arg \zeta(s)|\big).$$
Here the factor $1 + |\zeta(s)|$ avoids any lower-bound assumption on $|\zeta(s)|$. Proposition 3.4, applied with $\delta = \delta'$, gives $|H(s, y)| \ll_{\eta,\delta'} 1$. Moreover, Theorem 4.3 gives
$$|\zeta(s)| \ll_{\delta',\varepsilon} |t|^{\delta'/2+\varepsilon},$$
and Theorem 4.4 gives
$$|\arg \zeta(s)| \leq C_{\mathrm{arg}} \log |t|.$$
Since $|t| \geq 2$, these estimates imply
$$|F(s, y)| \ll_{\eta,\delta',\varepsilon} |t|^{(2+2\eta)(\delta'/2+\varepsilon)+2\eta C_{\mathrm{arg}}} = |t|^B,$$
which proves (4.28).

We now prove the low-height estimate. Let
$$q(s) := (s - 1)\zeta(s), \qquad q(1) := 1.$$
Then q is holomorphic at $s = 1$. Put
$$\lambda_0 := \frac{c_0}{2 \log 3}, \qquad K := \{s = \sigma + it : 1 - \lambda_0 \leq \sigma \leq 1,\ |t| \leq 2\}.$$
For $|t| \leq 2$ we have $\log(|t| + 2) \leq \log 4 < 2 \log 3$, and hence
$$\lambda_0 = \frac{c_0}{2 \log 3} < \frac{c_0}{\log(|t| + 2)}.$$
Thus $K \setminus \{1\}$ lies in the zero-free region of Theorem 4.2. After setting $q(1) = 1$, the function q is holomorphic and non-vanishing on K. Since K is compact, there is an open neighborhood U of K on which q is non-vanishing. Choose a holomorphic branch
$$M(s) = \log q(s)$$

on U, compatible with $M(s) = L(s) + \mathrm{Log}(s-1)$ on $U \cap \mathcal{D}$. Let

$$M_K := \sup_{s \in K} |M(s)| < \infty.$$

For $|y| \leq \eta$,

$$|q(s)^\alpha| = |\exp(\alpha M(s))| \leq \exp(|\alpha|\,|M(s)|) \leq \exp((2+2\eta)M_K) \qquad (s \in K).$$

Define

$$G(s,y) := q(s)^\alpha H(s,y). \tag{4.30}$$

Then, on the chosen branch near the slit,

$$F(s,y) = G(s,y)(s-1)^{-\alpha}. \tag{4.31}$$

Since $K \subset \{s : \mathrm{Re}(s) \geq 1 - \delta'\}$, combining the preceding bound for $q(s)^\alpha$ with Proposition 3.4, again with $\delta = \delta'$, gives

$$|G(s,y)| = |q(s)^\alpha H(s,y)| \ll_{\eta,\delta'} 1 \qquad (s \in K,\ |y| \leq \eta).$$

On the left vertical line $s = 1 - \lambda_T + it$ with $|t| \leq 2$, we use (4.31). For either boundary value of the logarithm along the slit, $|\mathrm{Arg}(s-1)| \leq \pi$, so

$$|(s-1)^{-\alpha}| = |s-1|^{-\mathrm{Re}(\alpha)} \exp\big(\Im(\alpha)\,\mathrm{Arg}(s-1)\big) \ll_\eta |s-1|^{-\mathrm{Re}(\alpha)}.$$

Since $|s-1| \geq \lambda_T$, if $|s-1| \leq 1$ then

$$|s-1|^{-\mathrm{Re}(\alpha)} \leq |s-1|^{-2-2\eta} \leq \lambda_T^{-2-2\eta}.$$

If $|s-1| \geq 1$, then $\mathrm{Re}(\alpha) \geq 2 - 2\eta > 0$, so $|s-1|^{-\mathrm{Re}(\alpha)} \leq 1 \leq \lambda_T^{-2-2\eta}$. Hence

$$|F(1-\lambda_T+it,y)| \ll_{\eta,\delta'} \lambda_T^{-2-2\eta}.$$

The definition (4.14) gives

$$\lambda_T^{-2-2\eta} = \left(\frac{2\log(T+2)}{c_0}\right)^{2+2\eta} \ll_{\eta,\delta'} (\log(T+2))^{2+2\eta},$$

which proves (4.29). $\square$

Since

$$B(\eta,\delta',\varepsilon) = (2+2\eta)\left(\tfrac{\delta'}{2}+\varepsilon\right) + 2\eta C_{\mathrm{arg}} \longrightarrow \delta' \qquad (\eta,\varepsilon \to 0^+),$$

and $\delta' < c_0/2$, we may choose $\eta > 0$ and $\varepsilon > 0$ sufficiently small so that

$$B < c_0/2. \tag{4.32}$$

This choice is fixed throughout the contour estimates.

Nonlocal contour pieces

Lemma 4.6 (Bound for the horizontal contour pieces)**.** For all sufficiently large x and every $T \geq 2$,

$$|I_{\rm top}| + |I_{\rm bot}| \ll_{\eta,\varepsilon} x^c\, T^{B-1}, \tag{4.33}$$

uniformly in $|y| \leq \eta$.

Proof. On $\mathcal{C}_{\rm top}$, parameterize $s = \sigma + iT$ with $\sigma_0(T) \leq \sigma \leq c$. Then

$$\begin{aligned}
|I_{\rm top}| &= \left| \frac{1}{2\pi i} \int_{\mathcal{C}_{\rm top}} F(s,y) \frac{x^s}{s}\, ds \right| \\
&\leq \frac{1}{2\pi} \int_{\sigma_0(T)}^{c} \left| F(\sigma+iT,y) \frac{x^{\sigma+iT}}{\sigma+iT} \right| d\sigma \\
&\leq \frac{c-\sigma_0(T)}{2\pi} \sup_{\sigma_0(T)\leq\sigma\leq c} \left| F(\sigma+iT,y) \frac{x^{\sigma+iT}}{\sigma+iT} \right|.
\end{aligned}$$

On this segment, $|x^{\sigma+iT}| = x^\sigma \leq x^c$ and $|\sigma+iT| \geq T$. Also $\sigma_0(T) \geq 1-\delta'$ by (4.15), while $c \leq 1+\delta'$ for all sufficiently large x. Hence Lemma 4.5 gives

$$\sup_{\sigma_0(T)\leq\sigma\leq c} |F(\sigma+iT,y)| \ll_{\eta,\varepsilon} T^B.$$

Since

$$c - \sigma_0(T) = \frac{1}{\log x} + \lambda_T = \frac{1}{\log x} + \frac{c_0}{2\log(T+2)} \ll 1,$$

we obtain

$$\begin{aligned}
|I_{\rm top}| &\ll_{\eta,\varepsilon} \frac{x^c}{T} \sup_{\sigma_0(T)\leq\sigma\leq c} |F(\sigma+iT,y)| \\
&\ll_{\eta,\varepsilon} x^c T^{B-1}.
\end{aligned}$$

The same computation on $\mathcal{C}_{\rm bot}$, with $s = \sigma - iT$, gives

$$|I_{\rm bot}| \ll_{\eta,\varepsilon} x^c T^{B-1}.$$

This proves (4.33). $\square$

Lemma 4.7 (Bound for the left vertical contour piece)**.** For every $T \geq 2$,

$$|I_{\rm left}| \ll_{\eta,\varepsilon} x^{1-\lambda_T}\, T^B, \tag{4.34}$$

uniformly in $|y| \leq \eta$.

Proof. The contour $\mathcal{C}_{\rm left} = \mathcal{C}^-_{\rm left} \cup \mathcal{C}^+_{\rm left}$ is estimated by parameterizing the two boundary-value pieces as $s = \sigma_0(T) + it$, with $-T \leq t < 0$ on $\mathcal{C}^-_{\rm left}$ and $0 < t \leq T$ on $\mathcal{C}^+_{\rm left}$. Using $\sigma_0(T) = 1 - \lambda_T$, we first write

$$\begin{aligned}
|I_{\rm left}| &= \left| \frac{1}{2\pi i} \int_{\mathcal{C}_{\rm left}} F(s,y) \frac{x^s}{s}\, ds \right| \\
&\leq \frac{1}{2\pi} \left(\int_{-T}^{0^-} \left| F(\sigma_0(T)+it,y) \frac{x^{\sigma_0(T)+it}}{\sigma_0(T)+it} \right| dt + \int_{0^+}^{T} \left| F(\sigma_0(T)+it,y) \frac{x^{\sigma_0(T)+it}}{\sigma_0(T)+it} \right| dt \right) \\
&= \frac{x^{1-\lambda_T}}{2\pi} \left(\int_{-T}^{0^-} \left| \frac{F(\sigma_0(T)+it,y)}{\sigma_0(T)+it} \right| dt + \int_{0^+}^{T} \left| \frac{F(\sigma_0(T)+it,y)}{\sigma_0(T)+it} \right| dt \right).
\end{aligned}$$

Since $\sigma_0(T) > 1-\delta' > 1/2$, we have $|\sigma_0(T)+it| \gg 1$ for $|t| \le 2$ and $|\sigma_0(T)+it| \gg |t|$ for $2 \le |t| \le T$. Hence

$$|I_{\mathrm{left}}| \ll x^{1-\lambda_T} \left(\int_{|t|\le 2} |F(\sigma_0(T)+it,y)|\,dt + \int_{2\le|t|\le T} \frac{|F(\sigma_0(T)+it,y)|}{|t|}\,dt \right).$$

On $|t| \le 2$, Lemma 4.5 gives

$$|F(\sigma_0(T)+it,y)| \ll_{\eta,\delta'} (\log(T+2))^{2+2\eta}.$$

On $2 \le |t| \le T$, the point $\sigma_0(T)+it$ lies in $\mathcal{D}$ as a boundary-value limit and satisfies $1-\delta' \le \sigma_0(T) \le 1+\delta'$, so Lemma 4.5 gives

$$|F(\sigma_0(T)+it,y)| \ll_{\eta,\varepsilon} |t|^B.$$

Therefore

$$\begin{aligned} |I_{\mathrm{left}}| &\ll_{\eta,\delta',\varepsilon} x^{1-\lambda_T} \left((\log(T+2))^{2+2\eta} + \int_{2\le|t|\le T} |t|^{B-1}\,dt \right) \\ &\ll_{\eta,\delta',\varepsilon} x^{1-\lambda_T} \left((\log(T+2))^{2+2\eta} + 2\int_2^T t^{B-1}\,dt \right). \end{aligned}$$

Since $B > 0$,

$$2\int_2^T t^{B-1}\,dt = \frac{2}{B}\bigl(T^B - 2^B\bigr) \ll_B T^B.$$

Also $(\log(T+2))^{2+2\eta} \ll_{\eta,B} T^B$ for $T \ge 2$. Combining these estimates gives

$$|I_{\mathrm{left}}| \ll_{\eta,\delta',\varepsilon} x^{1-\lambda_T} T^B.$$

This proves (4.34). □

Choice of T and the combined error

We choose

$$T = T(x) := \exp\bigl(\sqrt{\log x}\bigr), \tag{4.35}$$

the standard Selberg–Delange truncation height adapted to the de la Vallée Poussin zero-free region. With this choice, $\log T = \sqrt{\log x}$, and $\lambda_T = c_0/(2\sqrt{\log x} + O(1))$, so $\lambda_T < \delta'$ holds automatically. Moreover, $c = 1 + 1/\log x \le 1 + \delta'$ for all x sufficiently large, so the abscissa c lies inside the strip used in the growth estimates. We have

$$x^{-\lambda_T} = \exp(-\lambda_T \log x) = \exp\left(-\tfrac{c_0}{2}\sqrt{\log x} \cdot (1+o(1)) \right). \tag{4.36}$$

Lemma 4.8 (Short interval bound for d_K)**.** For each fixed integer $K \ge 1$, there exists a constant $C_K > 0$ such that, for all sufficiently large X, if

$$C_K X^{1-1/(2K)} \le H \le X,$$

then

$$\sum_{X<n\le X+H} d_K(n) \ll_K H(\log X)^{K-1}.$$

Here $d_K(n)$ is defined by

$$\zeta(s)^K = \sum_{n\ge1} d_K(n) n^{-s} \qquad (\mathrm{Re}(s) > 1),$$

or equivalently as the number of ordered K-tuples of positive integers whose product is n.

Proof. We prove the assertion by induction on K. For $K = 1$, we have $d_1(n) = 1$, so the estimate is immediate.

Assume $K \geq 2$ and that the result is known for $K - 1$. Put

$$D_K(X, H) := \sum_{X < n \leq X+H} d_K(n).$$

This counts ordered K-tuples $(m_1, \ldots, m_K)$ satisfying

$$X < m_1 \cdots m_K \leq X + H.$$

Since $H \leq X$, the product is at most $2X$, so at least one factor is at most $(2X)^{1/K}$. By the union bound over the K possible positions of such a factor,

$$D_K(X, H) \leq K \sum_{m \leq (2X)^{1/K}} \sum_{X/m < r \leq (X+H)/m} d_{K-1}(r).$$

Let $Y = X/m$ and $L = H/m$. Choose C_K large enough, in terms of K and C_{K-1}, so that whenever $H \geq C_K X^{1-1/(2K)}$ and $m \leq (2X)^{1/K}$, we have

$$L \geq C_{K-1} Y^{1-1/(2(K-1))}.$$

Also $L \leq Y$, since $H \leq X$. Hence the induction hypothesis applies to the inner sum and gives

$$\sum_{X/m < r \leq (X+H)/m} d_{K-1}(r) \ll_K \frac{H}{m} (\log X)^{K-2}.$$

Therefore

$$D_K(X, H) \ll_K H (\log X)^{K-2} \sum_{m \leq (2X)^{1/K}} \frac{1}{m} \ll_K H (\log X)^{K-1}.$$

□

Lemma 4.9 (Perron truncation error)**.** Let $T = T(x) = \exp(\sqrt{\log x})$ and $c = 1 + 1/\log x$. For $x \geq 2$ with $x \notin \mathbb{Z}$ and $|y| \leq \eta$, there exists a fixed integer $K = K(\eta)$ such that

$$E_{\text{Perron}}(x, T, y) \ll_\eta \frac{x}{T} (\log x)^K. \tag{4.37}$$

Proof. Set $M = 1 + \eta$ and choose an integer $K = K(\eta)$ with $K \geq 2M$ and $K \geq 2$. We first show that

$$d(n) M^{\omega(n)} \leq d_K(n) \qquad (n \geq 1).$$

For a prime power p^a with $a \geq 1$,

$$d(p^a) M^{\omega(p^a)} = (a+1)M, \qquad d_K(p^a) = \binom{a+K-1}{K-1}.$$

The quantity

$$\frac{1}{a+1} \binom{a+K-1}{K-1}$$

is increasing in a for $K \geq 2$, and at $a = 1$ it equals $K/2$. Thus $(a+1)M \leq d_K(p^a)$ for every $a \geq 1$. Extending multiplicatively gives the claimed coefficient domination.

Using (4.6), we obtain

$$E_{\text{Perron}}(x,T,y) \ll_\eta x^c \sum_{n\geq 1} \frac{d_K(n)}{n^c} \frac{1}{1+T\left|\log \frac{x}{n}\right|}.$$

We split the sum into the ranges $n \leq x/2$, $x/2 < n < 2x$, and $n \geq 2x$.

If $n \leq x/2$ or $n \geq 2x$, then $|\log(x/n)| \geq \log 2$, and hence the weight is $\ll 1/T$. The contribution of these two far ranges is therefore

$$\ll \frac{x^c}{T} \sum_{n\geq 1} \frac{d_K(n)}{n^c} = \frac{x^c}{T}\zeta(c)^K.$$

Since $c = 1 + 1/\log x$, we have $x^c \asymp x$ and $\zeta(c) \ll \log x$, so the far contribution is

$$\ll_K \frac{x}{T}(\log x)^K.$$

It remains to estimate the central range $x/2 < n < 2x$. There $x^c n^{-c} = (x/n)^c \ll 1$, so it is enough to bound

$$\sum_{x/2<n<2x} d_K(n) \frac{1}{1+T\left|\log \frac{x}{n}\right|}.$$

Put $H = x/T$. Cover the central range by the right intervals

$$I_r^+ = (x + rH,\ x + (r+1)H]$$

and the left intervals

$$I_r^- = [x - (r+1)H,\ x - rH),$$

with $r = 0, 1, 2, \ldots$ and only $O(T)$ intervals needed. For $n \in I_r^\pm$,

$$\frac{1}{1+T\left|\log \frac{x}{n}\right|} \ll \frac{1}{1+r}.$$

For $r = 0$ this is trivial. For $r \geq 1$, since $x/2 < n < 2x$, we have

$$\left|\log \frac{x}{n}\right| \asymp \frac{|n-x|}{x} \asymp \frac{rH}{x} = \frac{r}{T}.$$

Because $T = \exp(\sqrt{\log x})$, the length $H = x\exp(-\sqrt{\log x})$ is larger than $C_K x^{1-1/(2K)}$ for all sufficiently large x. Each interval lies in $x/2 < n < 2x$, so Lemma 4.8 applies with a starting point comparable to x, and gives

$$\sum_{n\in I_r^\pm} d_K(n) \ll_K H(\log x)^{K-1}.$$

Therefore the central contribution is

$$\ll_K H(\log x)^{K-1} \sum_{r\ll T} \frac{1}{1+r} \ll_K \frac{x}{T}(\log x)^{K-1}\log T.$$

Since $\log T = \sqrt{\log x} \leq \log x$ for sufficiently large x, this is

$$\ll_K \frac{x}{T}(\log x)^K.$$

Combining the far and central estimates proves (4.37). □

Proposition 4.10. With $T = T(x) = \exp(\sqrt{\log x})$, there exists a constant $c_1 > 0$ (depending on η, ε but not on x) such that

$$E_{\text{Perron}}(x, T, y) + |I_{\text{top}}| + |I_{\text{bot}}| + |I_{\text{left}}| \ll_{\eta,\varepsilon} x \exp\bigl(-c_1 \sqrt{\log x}\bigr), \tag{4.38}$$

uniformly in $|y| \le \eta$, for all sufficiently large $x \notin \mathbb{Z}$.

Proof. We bound each term separately and take the worst.

Perron error. By Lemma 4.9 with $T = \exp(\sqrt{\log x})$,

$$E_{\text{Perron}} \ll_\eta x \exp(-\sqrt{\log x})(\log x)^K.$$

Since $(\log x)^K = \exp(K \log\log x)$ and, for sufficiently large x,

$$K \log\log x \le \frac{1}{2}\sqrt{\log x},$$

we have

$$E_{\text{Perron}} \ll_\eta x \exp\left(-\frac{1}{2}\sqrt{\log x}\right).$$

Horizontal pieces. With $c = 1 + 1/\log x$ so that $x^c = e \cdot x$, by Lemma 4.6,

$$|I_{\text{top}}| + |I_{\text{bot}}| \ll_{\eta,\varepsilon} x \cdot T^{B-1} = x \cdot \exp\bigl((B-1)\sqrt{\log x}\bigr).$$

Since $B < 1$, this is $\ll x \exp(-(1-B)\sqrt{\log x})$.

Vertical piece. By Lemma 4.7 and (4.36),

$$|I_{\text{left}}| \ll_{\eta,\varepsilon} x \cdot x^{-\lambda_T} \cdot T^B = x \cdot \exp\bigl(-\tfrac{c_0}{2}\sqrt{\log x} + B\sqrt{\log x}\bigr)(1 + o(1)).$$

By the choice (4.13) of δ' together with (4.32), we have $B < c_0/2$. Hence $|I_{\text{left}}| \ll x \exp(-(c_0/2 - B)\sqrt{\log x})$.

Set

$$c_1 := \frac{1}{2}\min\left(\frac{1}{2},\ 1 - B,\ \frac{c_0}{2} - B\right) > 0.$$

For all sufficiently large x, the preceding three estimates are bounded by the right-hand side of (4.38). $\square$

The right-hand side of (4.38) is exponentially smaller than the local contribution on the $A(x, y)$ level. The local Hankel contribution I_H, analyzed in Section 5, contains the main term from which the asymptotic formula for each fixed $T_j(x)$ will later be extracted.

5 Local Hankel analysis

The purpose of this section is to analyze the local Hankel contribution I_H around the singular point $s = 1$. Recall that

$$F(s, y) = \zeta(s)^{2+2y} H(s, y).$$

The singular part is carried by the power of $\zeta(s)$; the remaining factor should be made regular at $s = 1$ before the local Hankel integral is evaluated.

5.1 The local regular factor

We keep the abbreviation

$$\alpha = 2 + 2y.$$

Define

$$q(s) := (s-1)\zeta(s), \qquad q(1) := 1. \tag{5.1}$$

The pole of $\zeta(s)$ at $s = 1$ is simple with residue 1, so q is holomorphic near $s = 1$ after filling in the removable singularity. After shrinking the neighborhood of $s = 1$ if necessary, we may assume that $q(s) \neq 0$ there. We then choose a single branch of $\log q(s)$ in this neighborhood, normalized by

$$\log q(1) = 0.$$

For s in this neighborhood and y near 0, define

$$G(s,y) := q(s)^{2+2y} H(s,y), \qquad q(s)^{2+2y} := \exp\big((2+2y)\log q(s)\big). \tag{5.2}$$

On the corresponding punctured slit neighborhood of $s = 1$, the branch choices from Section 4 give

$$F(s,y) = G(s,y)(s-1)^{-2-2y}. \tag{5.3}$$

Here the branch of $(s-1)^{-2-2y}$ is the local Hankel branch induced by the contour and branch conventions already fixed in Section 4.3.

Lemma 5.1 (Analyticity and boundedness of the local factor)**.** Fix $0 < \eta < 1/2$ and $0 < \delta < 1/2$. There exist $r \in (0,\delta)$ and $C_{\eta,\delta} > 0$ such that $G(s,y)$ is holomorphic in (s,y) for $|s-1| < r$ and $|y| < \eta$, and

$$|G(s,y)| \le C_{\eta,\delta}$$

uniformly for $|s-1| \le r$ and $|y| \le \eta$. Moreover, on the punctured slit neighborhood

$$0 < |s-1| < r, \qquad s \notin \mathcal{S},$$

we have the local factorization

$$F(s,y) = G(s,y)(s-1)^{-2-2y}.$$

Proof. Apply Proposition 3.4 with this fixed δ. It gives that the function $H(s,y)$ is holomorphic in (s,y) on the interior of $\mathcal{R}_{\delta,\eta}$ and uniformly bounded on $\mathcal{R}_{\delta,\eta}$. Choose $r \in (0,\delta)$. Then $|s-1| \le r$ implies $\mathrm{Re}(s) \ge 1 - r > 1 - \delta$, so

$$H(s,y) \ll_{\eta,\delta} 1 \qquad (|s-1| \le r,\ |y| \le \eta).$$

The function $q(s) = (s-1)\zeta(s)$ is holomorphic at $s = 1$ after setting $q(1) = 1$. Since $q(1) = 1 \neq 0$, we shrink r, preserving $r \in (0,\delta)$, so that q has no zeros on $|s-1| \le r$. Then a holomorphic branch of $\log q(s)$ exists on $|s-1| < r$, normalized by $\log q(1) = 0$. Since this branch is holomorphic on a neighborhood of the closed disk $|s-1| \le r$, it is bounded there. Therefore

$$q(s)^{2+2y} = \exp\big((2+2y)\log q(s)\big)$$

is uniformly bounded for $|s-1| \le r$ and $|y| \le \eta$. Combining this local bound with the bound for $H(s,y)$ gives $|G(s,y)| \le C_{\eta,\delta}$.

Finally, on the punctured slit neighborhood,

$$\zeta(s)^{2+2y} = q(s)^{2+2y}(s-1)^{-2-2y},$$

with the branch of $(s-1)^{-2-2y}$ inherited from the local Hankel contour. Combining this identity with $F(s,y) = \zeta(s)^{2+2y}H(s,y)$ proves the asserted factorization. $\square$

Lemma 5.2 (First-order approximation of the local factor)**.** Fix $0 < \eta < 1/2$ and $0 < \delta < 1/2$, and let $r \in (0, \delta)$ be chosen as in Lemma 5.1. Then

$$G(1, y) = H(1, y)$$

uniformly for $|y| \le \eta$, and

$$G(1, 0) = 1.$$

Moreover,

$$G(s, y) = G(1, y) + O_{\eta,\delta}(|s - 1|)$$

uniformly for $|s - 1| \le r/2$ and $|y| \le \eta$.

Proof. Since $q(1) = 1$ and the branch is normalized by $\log q(1) = 0$, we have

$$G(1, y) = q(1)^{2+2y} H(1, y) = H(1, y).$$

By Lemma 3.5, $H(1, 0) = 1$. Therefore $G(1, 0) = H(1, 0) = 1$.

For the first-order estimate, use Lemma 5.1. Since G is holomorphic in s on $|s - 1| < r$ and uniformly bounded for $|s - 1| \le r$, $|y| \le \eta$, Cauchy's estimate in the s-variable gives

$$\frac{\partial G}{\partial s}(s, y) \ll_{\eta,\delta} 1$$

uniformly for $|s - 1| \le r/2$ and $|y| \le \eta$. Therefore

$$G(s, y) - G(1, y) = \int_1^s \frac{\partial G}{\partial s}(w, y)\, dw = O_{\eta,\delta}(|s - 1|),$$

uniformly in the stated range. □

5.2 Evaluation of the local Hankel contribution

Proposition 5.3 (Local Hankel contribution)**.** With η fixed as in Section 4, put

$$T = \exp(\sqrt{\log x}), \qquad \rho := \frac{1}{\log x}.$$

For all sufficiently large x, uniformly for $|y| \le \eta$, we have

$$I_H = xH(1, y)\frac{(\log x)^{1+2y}}{\Gamma(2 + 2y)} + O_\eta\left(x(\log x)^{2\eta}\right).$$

Proof. Put $\alpha = 2+2y$. We apply Lemmas 5.1 and 5.2 with the fixed strip depth δ' from Section 4.3; since δ' is fixed, the constants below depend only on η. Let r be the local radius supplied by these lemmas. With $T = \exp(\sqrt{\log x})$, we have

$$\lambda_T = \frac{c_0}{2\log(T + 2)} \asymp \frac{1}{\sqrt{\log x}}.$$

Thus $\rho = 1/\log x < \lambda_T$ and $\rho < r/2$ for all sufficiently large x.

By the definition of I_H in (4.26), together with the local factorization recorded in Lemma 5.1,

$$\begin{aligned}I_H &:= \frac{1}{2\pi i}\int_{\mathcal{H}_{\rm loc}} F(s,y)\frac{x^s}{s}\,ds\\ &= \frac{1}{2\pi i}\int_{\mathcal{H}_{\rm loc}} G(s,y)(s-1)^{-\alpha}\frac{x^s}{s}\,ds\\ &= \frac{x}{2\pi i}\int_{\mathcal{H}_z} \frac{G(1+z,y)}{1+z} z^{-\alpha}e^{z\log x}\,dz,\end{aligned}$$

where the last equality follows from the change of variables $z = s-1$. Here $\mathcal{H}_z$ is the image of $\mathcal{H}_{\rm loc}$ under this change of variables. Thus $\mathcal{H}_z$ is the truncated Hankel contour whose arms run along the negative real axis from $-\lambda_T$ to $-\rho$, with the circle $|z| = \rho$ oriented counterclockwise through the right side of the origin.

For z on $\mathcal{H}_z$, Lemma 5.2 gives

$$G(1+z,y) = G(1,y) + O_\eta(|z|).$$

Also, since $|z| \le \lambda_T < 1/2$ for large x,

$$\frac{1}{1+z} = 1 + O(|z|).$$

Using the uniform boundedness of $G(1,y)$, we obtain

$$\frac{G(1+z,y)}{1+z} = G(1,y) + E(z,y), \qquad E(z,y) = O_\eta(|z|), \tag{5.4}$$

uniformly for $|y| \le \eta$ and $z \in \mathcal{H}_z$.

Accordingly, write

$$I_H = M_H + R_H,$$

where

$$M_H := \frac{xG(1,y)}{2\pi i}\int_{\mathcal{H}_z} z^{-\alpha}e^{z\log x}\,dz$$

and

$$R_H := \frac{x}{2\pi i}\int_{\mathcal{H}_z} E(z,y)z^{-\alpha}e^{z\log x}\,dz.$$

We first evaluate M_H. Put

$$U := \lambda_T \log x.$$

Under the change of variables $w = z\log x$, the contour $\mathcal{H}_z$ is mapped to a truncated Hankel contour $\mathcal{H}_U$ in the w-plane, whose arms run from $-U$ to -1 and whose circular part is $|w| = 1$. Then

$$M_H = xG(1,y)(\log x)^{\alpha-1}\frac{1}{2\pi i}\int_{\mathcal{H}_U} e^w w^{-\alpha}\,dw.$$

Let $\mathcal{H}_\infty$ be the corresponding infinite Hankel contour. By the standard Hankel representation of $1/\Gamma(\alpha)$ [],

$$\frac{1}{2\pi i}\int_{\mathcal{H}_\infty} e^w w^{-\alpha}\,dw = \frac{1}{\Gamma(\alpha)}.$$

The difference between $\mathcal{H}_\infty$ and $\mathcal{H}_U$ consists of the upper and lower tails over $(-\infty, -U]$. On these tails write $w = -u \pm i0$, with $u \geq U$. Then $|e^w| = e^{-u}$. Also, since $|y| \leq \eta$ and $\alpha = 2 + 2y$, the branch factors $e^{\pm i\pi\alpha}$ are bounded by a constant depending only on η, and hence

$$|w^{-\alpha}| \ll_\eta u^{-\operatorname{Re}(\alpha)}.$$

After taking absolute values, the two tails and the factor $1/(2\pi i)$ are absorbed into the O_η-constant, giving

$$\left| \frac{1}{2\pi i} \int_{\mathcal{H}_U} e^w w^{-\alpha}\, dw - \frac{1}{\Gamma(\alpha)} \right| \ll_\eta \int_U^\infty e^{-u} u^{-\operatorname{Re}(\alpha)}\, du.$$

Since

$$\operatorname{Re}(\alpha) = 2 + 2\operatorname{Re}(y) \geq 2 - 2\eta > 1$$

and $U \geq 1$ for sufficiently large x, we have

$$\int_U^\infty e^{-u} u^{-\operatorname{Re}(\alpha)}\, du \leq \int_U^\infty e^{-u}\, du = e^{-U}.$$

Therefore

$$\frac{1}{2\pi i} \int_{\mathcal{H}_U} e^w w^{-\alpha}\, dw = \frac{1}{\Gamma(\alpha)} + O_\eta(e^{-U}).$$

It follows that

$$M_H = xG(1, y)\frac{(\log x)^{\alpha-1}}{\Gamma(\alpha)} + O_\eta\left(x(\log x)^{\operatorname{Re}(\alpha)-1} e^{-U}\right).$$

Since $U = \lambda_T \log x \asymp \sqrt{\log x}$, there is a constant $c > 0$ such that $e^{-U} \ll e^{-c\sqrt{\log x}}$. Also $\operatorname{Re}(\alpha) - 1 \leq 1 + 2\eta$. Hence the tail contribution is

$$\ll_\eta x(\log x)^{1+2\eta} e^{-c\sqrt{\log x}} = O_\eta\left(x(\log x)^{2\eta}\right),$$

because $(\log x)e^{-c\sqrt{\log x}} = O(1)$.

It remains to estimate R_H. From (5.4),

$$|R_H| \ll_\eta x \int_{\mathcal{H}_z} |z|\, |z^{-\alpha}|\, e^{\operatorname{Re}(z)\log x}\, |dz|.$$

On the Hankel contour, $|\operatorname{Arg} z| \leq \pi$ and $|\Im\alpha| \leq 2\eta$, so

$$|z^{-\alpha}| \ll_\eta |z|^{-\operatorname{Re}(\alpha)}.$$

Thus

$$|R_H| \ll_\eta x \int_{\mathcal{H}_z} |z|^{1-\operatorname{Re}(\alpha)} e^{\operatorname{Re}(z)\log x}\, |dz|.$$

Changing variables again by $w = z\log x$ gives

$$|R_H| \ll_\eta x(\log x)^{\operatorname{Re}(\alpha)-2} \int_{\mathcal{H}_U} |w|^{1-\operatorname{Re}(\alpha)} e^{\operatorname{Re} w}\, |dw|.$$

The last integral is $O_\eta(1)$: this is immediate on $|w| = 1$, and on the arms it is bounded by

$$\int_1^U u^{1-\operatorname{Re}(\alpha)} e^{-u}\, du \leq \int_1^\infty u^{1-\operatorname{Re}(\alpha)} e^{-u}\, du \ll_\eta 1.$$

Consequently,

$$R_H = O_\eta \left(x(\log x)^{\mathrm{Re}(\alpha)-2} \right).$$

Since $\alpha = 2 + 2y$, we have

$$\mathrm{Re}(\alpha) - 2 = 2\,\mathrm{Re}(y) \le 2\eta,$$

and hence

$$R_H = O_\eta(x(\log x)^{2\eta}).$$

Combining the estimates for M_H and R_H, and using Lemma 5.2 to replace $G(1, y)$ by $H(1, y)$, we obtain

$$I_H = xH(1, y)\frac{(\log x)^{1+2y}}{\Gamma(2 + 2y)} + O_\eta(x(\log x)^{2\eta}),$$

uniformly for $|y| \le \eta$. □

Proposition 5.4 (Generating-function asymptotic)**.** With the parameters chosen as in Sections 4 and 5, and with the fixed contour parameters suppressed in the implied constant, uniformly for $|y| \le \eta$, as $x \to \infty$,

$$A(x, y) = xH(1, y)\frac{(\log x)^{1+2y}}{\Gamma(2 + 2y)} + O_\eta \left(x(\log x)^{2\eta} \right).$$

Proof. For non-integral x, the contour decomposition (4.24) gives

$$A(x, y) = I_H + I_{\mathrm{top}} + I_{\mathrm{bot}} + I_{\mathrm{left}} + E_{\mathrm{Perron}}(x, T, y).$$

By Proposition 4.10, for some $c_1 > 0$,

$$E_{\mathrm{Perron}}(x, T, y) + |I_{\mathrm{top}}| + |I_{\mathrm{bot}}| + |I_{\mathrm{left}}| \ll_\eta x \exp(-c_1\sqrt{\log x}),$$

uniformly for $|y| \le \eta$. Proposition 5.3 gives

$$I_H = xH(1, y)\frac{(\log x)^{1+2y}}{\Gamma(2 + 2y)} + O_\eta \left(x(\log x)^{2\eta} \right).$$

The exponentially small nonlocal and Perron error is absorbed into $O_\eta(x(\log x)^{2\eta})$. Therefore the asserted asymptotic holds for non-integral x.

For integral x, the replacement by $x + 1/2$ described in Section 4.1 gives the same estimate, since $A(x, y) = A(x + 1/2, y)$ and replacing $x + 1/2$ by x changes the displayed main term by an amount absorbed into the same error term. □

6 Extraction of $T_j(x)$ and proof of the main theorem

Proof of Theorem 1.3. By Proposition 5.4, uniformly for $|y| \le \eta$,

$$A(x, y) = xH(1, y)\frac{(\log x)^{1+2y}}{\Gamma(2 + 2y)} + O_\eta \left(x(\log x)^{2\eta} \right).$$

Define

$$K(y) := \frac{H(1, y)}{\Gamma(2 + 2y)}.$$

Then K is holomorphic in a neighborhood of 0, and

$$A(x,y) = x\log x\, K(y) e^{2y\log\log x} + O_\eta\left(x(\log x)^{2\eta}\right).$$

The generating relation

$$A(x,y) = \sum_{j\geq 0} T_j(x) y^j$$

gives the coefficient identity below. Here $[y^j]$ denotes the coefficient of y^j in the power series expansion:

$$T_j(x) = [y^j] A(x,y).$$

Write

$$K(y) = \sum_{r\geq 0} k_r y^r, \qquad k_r = \frac{K^{(r)}(0)}{r!},$$

and

$$e^{2y\log\log x} = \sum_{m\geq 0} \frac{(2\log\log x)^m}{m!} y^m.$$

For fixed j,

$$[y^j]\left(K(y)e^{2y\log\log x}\right) = \sum_{r=0}^{j} k_r \frac{(2\log\log x)^{j-r}}{(j-r)!}.$$

Therefore

$$[y^j]\left(K(y)e^{2y\log\log x}\right) = k_0 \frac{(2\log\log x)^j}{j!} + k_1 \frac{(2\log\log x)^{j-1}}{(j-1)!} + O_j\left((\log\log x)^{j-2}\right).$$

It remains to compute k_0 and k_1. By Lemma 3.5, $H(1,0) = 1$. Also $\Gamma(2) = 1$. Hence

$$k_0 = K(0) = 1.$$

Next,

$$k_1 = K'(0) = \partial_y H(1,0) + H(1,0)\, \frac{d}{dy} \frac{1}{\Gamma(2+2y)}\bigg|_{y=0}.$$

Here $\psi = \Gamma'/\Gamma$ denotes the digamma function. Since

$$\frac{d}{dy} \frac{1}{\Gamma(2+2y)}\bigg|_{y=0} = -2\psi(2) = -2(1-\gamma),$$

we get

$$k_1 = \partial_y H(1,0) - 2(1-\gamma).$$

We also need to extract coefficients from the error term. Choose a fixed radius ρ_y with $0 < \rho_y < \eta$. Since the error in Proposition 5.4 is uniform for $|y| \leq \eta$, Cauchy's coefficient estimate on $|y| = \rho_y$ gives, for each fixed j,

$$[y^j]\, O_\eta\left(x(\log x)^{2\eta}\right) = O_{\eta,j,\rho_y}\left(x(\log x)^{2\eta}\right).$$

Because $0 < \eta < 1/2$, for fixed $j \geq 2$,

$$x(\log x)^{2\eta} = O_j\left(x\log x(\log\log x)^{j-2}\right).$$

Thus this error is absorbed into the final error term.

Combining the preceding estimates gives, for each fixed $j \geq 2$,

$$\begin{aligned}T_j(x) = &\frac{2^j}{j!} x \log x (\log\log x)^j \\ &+ \frac{2^{j-1}}{(j-1)!} \left(\partial_y H(1,0) - 2(1-\gamma)\right) x \log x (\log\log x)^{j-1} \\ &+ O_j\left(x \log x (\log\log x)^{j-2}\right).\end{aligned}$$

In particular,

$$T_j(x) \sim \frac{2^j}{j!} x \log x (\log\log x)^j.$$

□

7 Toward $D(x)$: outlook

The identity (1.6) expresses $D(x)$ exactly in terms of the prime-power contribution and the alternating finite sum of the quantities $T_j(x)$. Theorem 1.3, however, is a fixed-j asymptotic statement. Since the range of relevant j grows with x, these fixed-j asymptotics cannot be inserted term by term into (1.6) to obtain an asymptotic formula for $D(x)$.

Equivalently, if

$$A(x,y) = \sum_{j \geq 0} T_j(x) y^j,$$

then the alternating part of (1.6) is determined by values of $A(x,y)$ and its y-derivative at $y = -1$. The analytic work in this paper is carried out for y in a small neighborhood of 0. Extending the method toward $D(x)$ would therefore require estimates uniform in j, or a direct analysis of the generating function near $y = -1$.

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